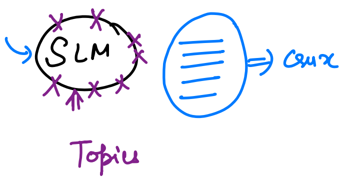
Agenda → Relation b/w words in a sentence

Review

Amazon → product → Review summary



Musical Instruments

Guitars

1.
2.
3.

Amp → 1000

Numpy of NLP → NLTK

Spacy → Natural language

frequent word Analysis

What exactly about guitars people are mentioning ?

Ans NO

↳ phrases , syntactic context

Spacy → 1 Min

Noun → parent

"Strings" → Strings are great / noisy / rusty.

noun

"Pedal" → Pedal is noisy / excellent

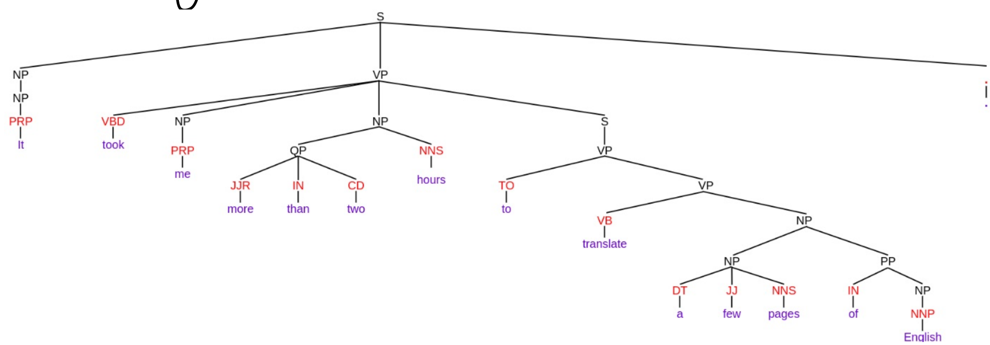
noun phrases →

How these nouns are mentioned

String are

Pedal is

Constituency Parsing



Dependency Parsing

https://demos.explosion.ai/displacy

Topic Modelling

LDA

Key Steps to Implement LDA

To run LDA in Gensim, you typically follow a specific pipeline.

Pre-processing: Clean your text (remove stopwords, punctuation, and lemmatize words).

Dictionary Creation: Use the Gensim Dictionary to map every unique word to an ID.

Corpus Creation: Convert your documents into a **Bag-of-Words (BoW)** format (a list of `(word_id, word_count)` tuples).

Topic 0:
`"0.020*"great" + 0.014*"good" + 0.012*"sound" + 0.012*"strings" + ..."`

Topic 1:
`"0.013*"guitar" + 0.010*"one" + 0.009*"case" + 0.008*"bag" + ..."`

Topic 2:
`"0.011*"amp" + 0.010*"stand" + 0.010*"sound" + 0.009*"use" + ..."`

Topic 5:
`"0.032*"strings" + 0.026*"guitar" + 0.012*"tuner" + 0.011*"tune" + ..."`

Topic 9:
`"0.021*"pedal" + 0.012*"use" + 0.012*"one" + 0.011*"good" + 0.011*"like" + ..."`

Topic 5 → "Strings & Tuning"

Topic 2 → "Amps & Stands"

Topic 9 → "Pedals & Tone"

```
1. Raw Amazon review:
   "These strings are bright and stay in tune for months on my acoustic guitar."

2. Basic tokenization (with NLTK):
   ["These", "strings", "are", "bright", "and", "stay", "in", "tune",
    "for", "months", "on", "my", "acoustic", "guitar", "."]

3. Stopword removal:
   ["strings", "bright", "stay", "tune", "months", "acoustic", "guitar"]

4. POS tagging + noun chunks (spaCy):
   Nouns/PROPN: "strings", "tune", "months", "guitar"
   Noun phrases: ["strings", "tune", "acoustic guitar"]

5. Noun phrase + parent:
   "strings->are", "acoustic guitar->on"

6. Topic modelling input (LDA) from cleaned tokens:
   ["strings", "bright", "stay", "tune", "months", "acoustic", "guitar"]

7. LDA topic assignment:
   Document is 70% Topic 5 ("Strings & Tuning"), 20% Topic 1 ("Guitars & Cases"), 10%
others.

8. Interpretation:
   - Customers often talk about strings, tuning, and acoustic guitars.
   - The topics show “Strings & Tuning” as a major recurring theme.
```